

# SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

## Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

## Building Information

- Weather File: USA\_TX\_Amarillo-Husband.Amarillo.Intl.AP.723630\_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m<sup>2</sup>

## Renovation Details by Scenario

| Scenario   | Renovation Type | Renovation Details                                                                                                                                                                       |
|------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline   | baseline        | Baseline (no envelope renovation)                                                                                                                                                        |
| Scenario 1 | wall + roof     | <b>Wall upgrade:</b> Exterior walls were upgraded to R-10.0 with Fiberglass Batts insulation.<br><b>Roof upgrade:</b> The roof was upgraded to R-20.0 using Blown Fiberglass insulation. |
| Scenario 2 | wall + roof     | <b>Wall upgrade:</b> Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation.                                                                                            |

|            |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                             | <b>Roof upgrade:</b> The roof was upgraded to R-24.4 using Blown Fiberglass insulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Scenario 3 | wall + roof                 | <p><b>Wall upgrade:</b> Exterior walls were upgraded to R-13.0 with Expanded Polystyrene (EPS) Foam Board insulation.</p> <p><b>Roof upgrade:</b> The roof was upgraded to R-24.4 using Extruded Polystyrene (XPS) Foam Board insulation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Scenario 4 | wall + roof                 | <p><b>Wall upgrade:</b> Exterior walls were upgraded to R-20.0 with Fiberglass Batts insulation.</p> <p><b>Roof upgrade:</b> The roof was upgraded to R-30.0 using Blown Fiberglass insulation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Scenario 5 | wall + roof + window + door | <p><b>Wall upgrade:</b> Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation.</p> <p><b>Roof upgrade:</b> The roof was upgraded to R-24.4 using Blown Fiberglass insulation.</p> <p><b>Window upgrade:</b></p> <ul style="list-style-type: none"> <li>• Upgraded to 3-pane glazing</li> <li>• Weatherstripping: silicone adhesive smoke gasket</li> <li>• Glazing film: low-e film</li> <li>• Window frame: wood window frame</li> <li>• Caulking: acrylic</li> <li>• Infiltration reduction: 30%</li> </ul> <p><b>Door upgrade:</b></p> <ul style="list-style-type: none"> <li>• Door type: wooden door</li> <li>• Bottom seal: automatic door bottom</li> <li>• Top/side seals: jamb weatherstrip</li> <li>• Infiltration reduction: 30%</li> </ul> |

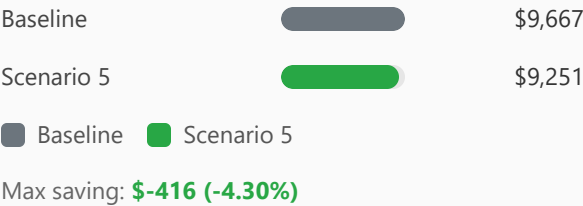
## Annual Energy Analysis

| Scenario | Metric | Baseline | Renovation | Delta | Savings % |
|----------|--------|----------|------------|-------|-----------|
|----------|--------|----------|------------|-------|-----------|

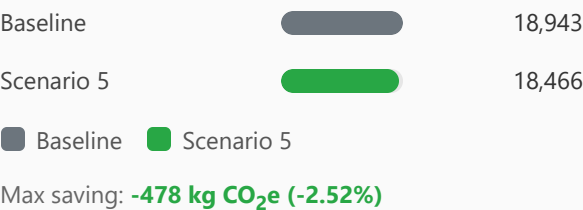
|            |                        |        |        |      |       |
|------------|------------------------|--------|--------|------|-------|
| Scenario 1 | Total Site Energy (GJ) | 195.48 | 193.42 | 2.06 | 1.05% |
| Scenario 2 | Total Site Energy (GJ) | 195.48 | 192.49 | 2.99 | 1.53% |
| Scenario 3 | Total Site Energy (GJ) | 195.48 | 192.49 | 2.99 | 1.53% |
| Scenario 4 | Total Site Energy (GJ) | 195.48 | 191.34 | 4.14 | 2.12% |
| Scenario 5 | Total Site Energy (GJ) | 195.48 | 190.41 | 5.07 | 2.59% |

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO<sub>2</sub>e/yr) Comparison



Min Retrofit Construction Cost

\$45,576

Scenario 1

Min Retrofit Embodied Carbon

5,558 kg CO<sub>2</sub>e

Embodied carbon intensity (ECI): 11 kg CO<sub>2</sub>e/m<sup>2</sup>  
Scenario 1

Lowest Retrofit Cost Payback Period

213 years

Scenario 4

Lowest Retrofit Carbon Payback Period

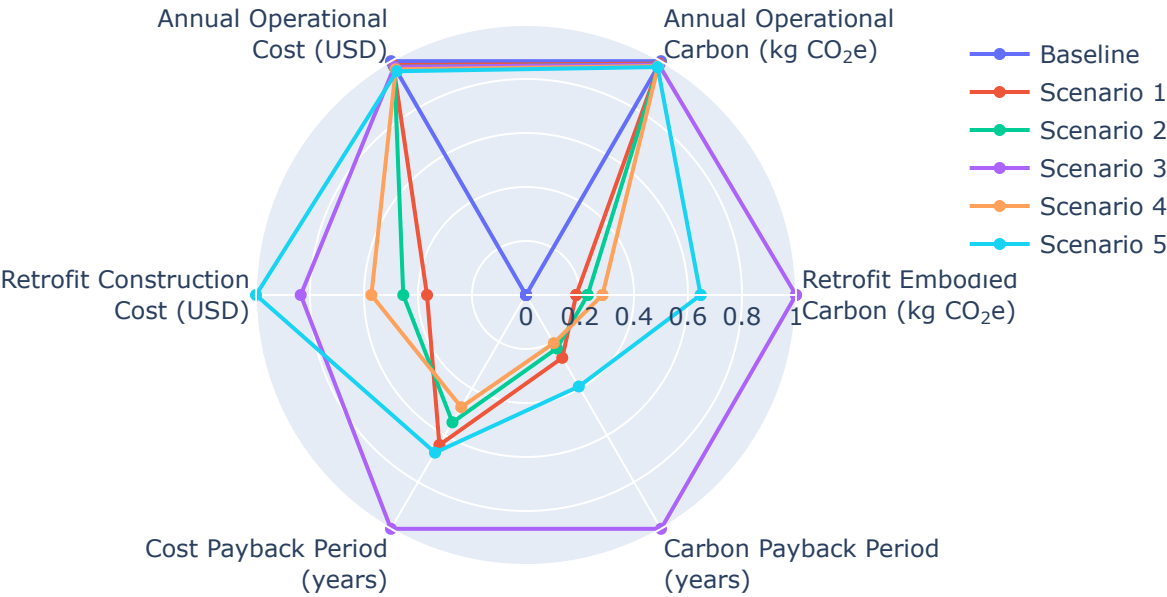
21 years

Scenario 4

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

|            |                        |         |
|------------|------------------------|---------|
| Scenario 1 | <div><div></div></div> | 284 yrs |
| Scenario 2 | <div><div></div></div> | 241 yrs |
| Scenario 3 | <div><div></div></div> | 443 yrs |
| Scenario 4 | <div><div></div></div> | 213 yrs |
| Scenario 5 | <div><div></div></div> | 299 yrs |

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO<sub>2</sub>e/yr)).

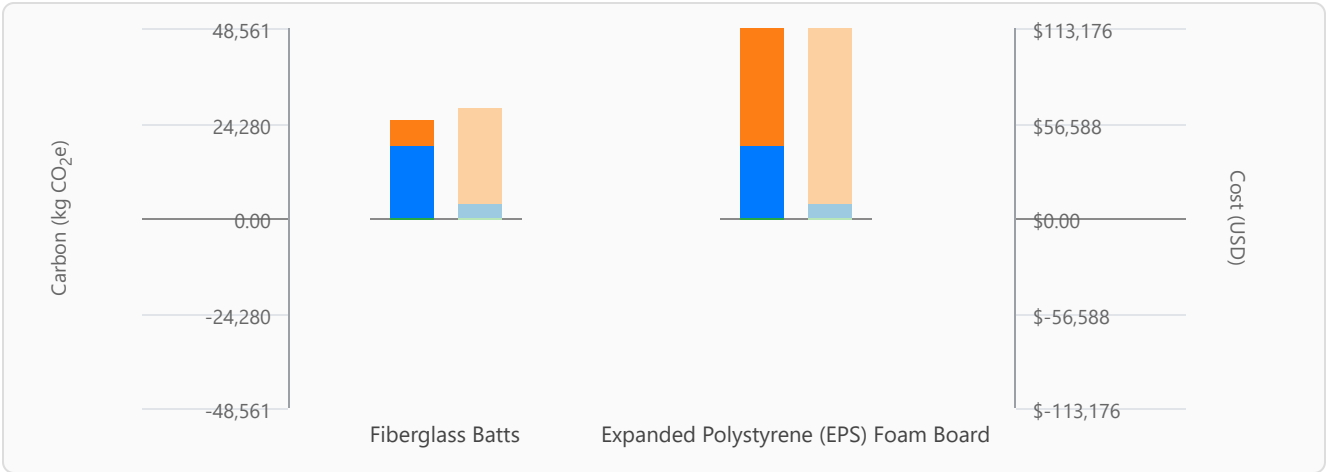
|            |                        |         |
|------------|------------------------|---------|
| Scenario 1 | <div><div></div></div> | 28 yrs  |
| Scenario 2 | <div><div></div></div> | 24 yrs  |
| Scenario 3 | <div><div></div></div> | 104 yrs |
| Scenario 4 | <div><div></div></div> | 21 yrs  |
| Scenario 5 | <div><div></div></div> | 41 yrs  |

Material Type Comparison

Auto-generated when scenarios share the same target insulation settings but use different insulation materials.

Wall insulation material comparison (target R-13.0)

| Scenario   | Insulation Material Type              | Operational Cost Saving (Scenario - Baseline) (\$/yr) | Operational Carbon Saving (Scenario - Baseline) (kg CO <sub>2</sub> e/yr) | Cost Payback Period (years) | Carbon Payback Period (years) |
|------------|---------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|-------------------------------|
| Scenario 2 | Fiberglass Batts                      | \$-234                                                | -289                                                                      | 241 yrs                     | 24 yrs                        |
| Scenario 3 | Expanded Polystyrene (EPS) Foam Board | \$-234                                                | -289                                                                      | 443 yrs                     | 104 yrs                       |



- Annual operational carbon (positive)

■ Retrofit embodied carbon (positive)

■ Annual operational carbon saving (negative)
- Annual operational cost (positive)

■ Retrofit construction cost (positive)

■ Annual operational cost saving (negative)

## Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

| Scenario   | Component       | Material         | Volume (m <sup>3</sup> ) | Area (m <sup>2</sup> ) | Thickness (m) | Length (m) | Thermal Conductivity (W/m·K) | Density (kg/m <sup>3</sup> ) |
|------------|-----------------|------------------|--------------------------|------------------------|---------------|------------|------------------------------|------------------------------|
| Scenario 1 | Wall insulation | Fiberglass Batts | N/A                      | 282                    | N/A           | N/A        | N/A                          | N/A                          |

|            |                         |                                       |     |     |     |      |     |     |
|------------|-------------------------|---------------------------------------|-----|-----|-----|------|-----|-----|
| Scenario 1 | Roof insulation         | Blown Fiberglass                      | N/A | 599 | N/A | N/A  | N/A | N/A |
| Scenario 2 | Wall insulation         | Fiberglass Batts                      | N/A | 282 | N/A | N/A  | N/A | N/A |
| Scenario 2 | Roof insulation         | Blown Fiberglass                      | N/A | 599 | N/A | N/A  | N/A | N/A |
| Scenario 3 | Wall insulation         | Expanded Polystyrene (EPS) Foam Board | N/A | 282 | N/A | N/A  | N/A | N/A |
| Scenario 3 | Roof insulation         | Extruded Polystyrene (XPS) Foam Board | N/A | 599 | N/A | N/A  | N/A | N/A |
| Scenario 4 | Wall insulation         | Fiberglass Batts                      | N/A | 282 | N/A | N/A  | N/A | N/A |
| Scenario 4 | Roof insulation         | Blown Fiberglass                      | N/A | 599 | N/A | N/A  | N/A | N/A |
| Scenario 5 | Wall insulation         | Fiberglass Batts                      | N/A | 282 | N/A | N/A  | N/A | N/A |
| Scenario 5 | Roof insulation         | Blown Fiberglass                      | N/A | 599 | N/A | N/A  | N/A | N/A |
| Scenario 5 | Window weatherstripping | silicone adhesive smoke gasket        | N/A | N/A | N/A | N/A  | N/A | N/A |
| Scenario 5 | Window caulking         | acrylic                               | N/A | N/A | N/A | N/A  | N/A | N/A |
| Scenario 5 | Door bottom seal        | automatic door bottom                 | N/A | N/A | N/A | 3.66 | N/A | N/A |
| Scenario 5 | Door top/side seal      | jamb weatherstripping                 | N/A | N/A | N/A | 16   | N/A | N/A |

## Result Summary

| Scenario   | Retrofit Embodied Carbon (kg CO <sub>2</sub> e) | Annual Operational Carbon (kg CO <sub>2</sub> e) | Annual Operational Cost (USD) | Retrofit Construction Cost (USD) |
|------------|-------------------------------------------------|--------------------------------------------------|-------------------------------|----------------------------------|
| Baseline   | 0.00                                            | 18,943                                           | \$9,667                       | \$0.00                           |
| Scenario 1 | 5,558                                           | 18,743                                           | \$9,507                       | \$45,576                         |
| Scenario 2 | 6,829                                           | 18,654                                           | \$9,433                       | \$56,511                         |
| Scenario 3 | 29,907                                          | 18,654                                           | \$9,433                       | \$103,743                        |
| Scenario 4 | 8,470                                           | 18,546                                           | \$9,333                       | \$71,196                         |
| Scenario 5 | 19,345                                          | 18,466                                           | \$9,251                       | \$124,324                        |

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